

SI2301

Si2301 — P-Channel 20 V MOSFET

Manufacturer: Vishay Package: SOT-23

General Description

The Si2301 is a P-channel enhancement mode MOSFET with a 20 V drain-source rating and 2.3 A continuous drain current in a SOT-23 package. It is used as a load switch in battery powered equipment.

Features

- Drain-source voltage $V_{DS} = -20\text{ V}$
- Continuous drain current $I_D = -2.3\text{ A}$
- Low on-resistance $R_{DS(on)} = 90\text{ milliohm}$ at $V_{GS} = -4.5\text{ V}$
- Logic level gate drive

Pin Configuration

Pin assignment for the SOT-23 package (top view).

Pin	Name	Function
1	Gate	Control input, active low
2	Source	Source, connect to supply
3	Drain	Switched output

Absolute Maximum Ratings

Stresses beyond these values may cause permanent damage to the device.

Symbol	Parameter	Value	Unit
V_{DS}	Drain-source voltage	-20	V
V_{GS}	Gate-source voltage	± 8	V
I_D	Continuous drain current	-2.3	A
I_{DM}	Pulsed drain current	-10	A
P_{tot}	Total power dissipation	750	mW

Electrical Characteristics

Typical values at TA = 25 degC unless otherwise noted.

Symbol	Parameter	Test condition	Min	Typ	Max	Unit
VGS(th)	Gate threshold voltage	VDS = VGS, ID = -250 uA	-0.45	-0.75	-1.2	V
RDS(on)	Static drain-source on-resistance	VGS = -4.5 V, ID = -2.3 A	-	70	90	milliohm
IDSS	Zero gate voltage drain current	VDS = -16 V	-	-	-1	uA
Qg	Total gate charge	VDS = -10 V, VGS = -4.5 V	-	5.5	9.0	nC

Thermal Characteristics

Symbol	Parameter	Value	Unit
RthJA	Thermal resistance junction-ambient	166	K/W

Package Dimensions

Outline drawing for the SOT-23 package. All dimensions in millimetres.

Dimension	Min	Nom	Max	Unit
A (body length)	2.80	2.90	3.04	mm
B (body width)	1.20	1.30	1.45	mm

Ordering Information

Order code	Package	Marking	Packing
SI2301BDS	SOT-23	A1	Reel, 3000 pcs

Document revision 1.0. Values are typical at TA = 25 degC. Verify against the latest datasheet from Vishay before design release.